Cardiovascular System (CARD) · NPLEX-P1-CARD-14

CV-14 Cardiovascular Genetics

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-14-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the LDL receptor pathway and its feedback

A competent response contains

· Each LDL particle carries one apolipoprotein B-100, which is the ligand for the LDL receptor.

· The liver clears about 70% of plasma LDL, so hepatic receptor number sets the plasma LDL concentration.

· Sequence: apoB-100 binds the receptor, the complex is internalized in clathrin-coated pits, the ligand is released in the acidified endosome while the receptor recycles to the surface, and LDL is hydrolyzed in the lysosome to free cholesterol.

· The receptor also binds apolipoprotein E on intermediate-density lipoprotein, so scarce receptors leave more IDL to become LDL.

· When endoplasmic reticulum cholesterol is low, SCAP escorts SREBP-2 to the Golgi, where proteases release the active fragment.

· The active SREBP-2 fragment switches on HMG-CoA reductase, the LDL receptor, and PCSK9.

· When endoplasmic reticulum cholesterol is high, INSIG anchors SREBP-2 so it never reaches the Golgi.

· PCSK9 is secreted by hepatocytes, binds the receptor, and directs it to the lysosome instead of letting it recycle, so PCSK9 activity sets how many trips each receptor makes.

· HMG-CoA reductase is the rate-limiting enzyme of cholesterol synthesis.

Distinctions that matter

· Receptor recycling and receptor degradation are different fates for the same protein, and PCSK9 decides between them.

· The receptor has two ligands: apoB-100 on LDL and apoE on IDL, which is why defects in the two proteins behave differently.

Common misconceptions

· Treating SREBP-2 as responding to plasma cholesterol. It responds to cholesterol in the endoplasmic reticulum membrane, which is why plasma can be loaded while the pathway stays switched on.

· Saying the receptor is consumed with each LDL particle. It is released in the endosome and recycles unless PCSK9 sends it to the lysosome.

Where it is taught: Section C01, first two paragraphs of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C01-BD2

Brain dump 2 · integration and application: Why plasma LDL stays high while the liver cell starves

A competent response contains

· Familial hypercholesterolemia is autosomal dominant, with heterozygotes about 1 in 250.

· Locus heterogeneity: loss-of-function LDLR variants on chromosome 19p in most cases, APOB variants in the receptor-binding region, and gain-of-function PCSK9 variants, which are the rarest.

· Familial defective apoB-100 is usually milder because apoE-mediated clearance of IDL is preserved.

· Heterozygotes have LDL cholesterol about twice normal from birth, usually above about 190 mg/dL (4.9 mmol/L) in untreated adults against a desirable value below about 100 mg/dL (2.6 mmol/L), with normal triglycerides, which is Fredrickson type IIa.

· Little LDL reaches the hepatocyte endoplasmic reticulum, so SREBP-2 stays active, synthesis continues, and VLDL secretion feeds future LDL.

· Reduced receptor number lowers clearance of both LDL and IDL, and the surviving IDL becomes LDL, so production rises as clearance falls.

· Cholesterol delivered by a route that bypasses the receptor suppresses synthesis normally in familial hypercholesterolemia fibroblasts, which proves that uptake is defective and feedback is intact.

· Lowering hepatic synthesis upregulates the normal allele in a heterozygote, which is why it works there and does little in a receptor-negative homozygote.

· Macrophage scavenger receptors take up oxidized LDL and are not downregulated, so foam cells form, producing xanthomas, corneal arcus, and premature atherosclerosis.

· Tendon xanthomas at the Achilles and hand extensors and corneal arcus before about age 45 fit the heterozygous phenotype, with coronary disease often by the fourth or fifth decade in men.

Distinctions that matter

· Defective uptake against defective feedback: the fibroblast experiment separates them, and only uptake is broken.

· Gain of function in PCSK9 raises LDL while loss of function in PCSK9 gives lifelong low LDL and markedly lower coronary risk, so the direction of the functional change matters more than the gene name.

Common misconceptions

· Saying the high plasma LDL should switch off hepatic synthesis. The hepatocyte cannot sense cholesterol it never takes up.

· Expecting the receptor pathway to limit foam cell formation. Scavenger receptors on macrophages are not downregulated by cell cholesterol.

Where it is taught: Section C01, paragraphs on familial hypercholesterolemia and on lost feedback of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C01-RT01

Retrieval exercise: Rebuild the familial hypercholesterolemia sequence

A competent response contains

· Step 1: an LDLR, APOB, or gain-of-function PCSK9 variant.

· Step 2: fewer working receptor trips at the hepatocyte surface.

· Step 3: less LDL and IDL cleared, and more IDL becomes LDL.

· Step 4: low endoplasmic reticulum cholesterol keeps SREBP-2 active and synthesis high.

· Step 5: plasma LDL is high from birth.

· Step 6: foam cells give xanthomas, corneal arcus, and premature atherosclerosis.

· Fewer receptors raise production as well as lowering clearance, because the receptor also clears IDL and unretrieved IDL is converted to LDL.

· SREBP-2 stays active and keeps switching on HMG-CoA reductase, the rate-limiting enzyme of cholesterol synthesis.

· The five classes of LDL receptor variant fail at synthesis, transport from endoplasmic reticulum to Golgi, ligand binding, clustering in coated pits, and recycling.

· Transport from endoplasmic reticulum to Golgi is the most common class.

Distinctions that matter

· Reduced clearance and continued production are two separate contributions to the same high LDL, and both follow from the one receptor defect.

· The five variant classes are defined by the step in the receptor's life cycle that fails, not by severity.

Common misconceptions

· Writing the sequence as a single clearance problem. Production stays high because feedback never sees the cholesterol.

· Placing the lipid deposits before the lifelong elevation. Deposits in skin, tendon, cornea, and artery follow decades of high LDL.

Where it is taught: Section C01, flow diagram on pathogenesis of familial hypercholesterolemia of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C02-BD1

Brain dump 1 · foundational retrieval: Sarcomere genes to the hypertrophic cardiomyopathy phenotype

A competent response contains

· Hypertrophic cardiomyopathy is left ventricular thickening not explained by abnormal loading such as hypertension or aortic stenosis.

· It affects about 1 in 500 adults and in most familial cases is an autosomal dominant disease of the sarcomere.

· MYH7 encodes beta-myosin heavy chain and MYBPC3 encodes cardiac myosin-binding protein C, and together they account for most genotype-positive cases.

· Less common genes are troponin T and troponin I, alpha-tropomyosin, the myosin light chains, and cardiac actin.

· MYH7 missense variants make a mutant myosin head built into the thick filament beside normal myosin, which is a dominant negative effect.

· MYBPC3 variants are usually truncating, and nonsense-mediated decay lowers the amount of protein, which is haploinsufficiency, with later and milder disease on average.

· Testing finds a variant in only about a third to a half of patients, so a negative test does not exclude a genetic cause.

· Sequence: variant, then more myosin heads cycling or higher Ca2+ sensitivity, then hypercontractility with high ATP use and impaired relaxation, then an energy deficit that activates hypertrophic and fibrotic signaling, then hypertrophy with myocyte disarray, fibrosis, and small vessel disease, then a stiff nondilated ventricle with an arrhythmic substrate.

· Many variants release myosin heads from their energy-saving folded resting state, and thin filament variants raise Ca2+ sensitivity.

· Adult wall thickness of about 15 mm or more is diagnostic, against about 11 mm or less normally, often with asymmetric septal hypertrophy and a small cavity, and the ventricle is not dilated.

· The hallmark is myocyte disarray, with short branched myocytes in whorls, accompanied by interstitial fibrosis and thick-walled intramural arteries.

Distinctions that matter

· Dominant negative against haploinsufficiency: MYH7 puts a faulty protein into the filament, while MYBPC3 simply supplies less protein, which is why MYBPC3 disease tends to be later and milder.

· Hypertrophy here follows from the motor itself, not from load, which is what separates it from hypertensive or valvular thickening.

Common misconceptions

· Describing the ventricle as weak. It is hypercontractile and relaxes poorly, and the problem is diastolic.

· Taking a negative gene panel as evidence against the diagnosis. Only about a third to a half of patients have a detectable variant.

Where it is taught: Section C02, first three paragraphs and the flow diagram of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C02-BD2

Brain dump 2 · integration and application: Two relatives, one variant, and a murmur that moves

A competent response contains

· Each child of a carrier has a 50% risk because the disease is autosomal dominant.

· Relatives with the same variant range from sudden death to a normal echocardiogram, which is variable expressivity with incomplete penetrance.

· Penetrance is age dependent, with hypertrophy usually appearing in adolescence or early adulthood, so a genotype-positive relative with a normal study needs repeated evaluation over time.

· With MYBPC3 hypertrophy sometimes does not appear until middle age.

· A second sarcomere variant raises severity, and male sex, hypertension, and modifier genes raise penetrance.

· The stiff ventricle fills at high pressure, which causes exertional dyspnea and a fourth heart sound.

· Filling depends on atrial contraction, so atrial fibrillation is poorly tolerated.

· About a third of patients have dynamic outflow obstruction at rest: the septum narrows the outflow tract and flow drags the anterior mitral leaflet toward it, which is systolic anterior motion, adding mitral regurgitation.

· Standing and the strain of a Valsalva maneuver reduce ventricular volume, so obstruction and the murmur increase; squatting, leg raising, and handgrip raise volume or afterload, so the murmur softens.

· The murmur of aortic stenosis responds in the opposite direction to the same volume changes.

· Angina occurs with normal coronary arteries because demand is high and the intramural vessels are narrow.

· Phenocopies: athlete's heart with an enlarged cavity, normal relaxation, and regression with detraining; hypertensive hypertrophy; Fabry disease from X-linked GLA; Friedreich ataxia; and Noonan syndrome.

Distinctions that matter

· Penetrance is whether any disease appears and expressivity is how severe it is; an age-dependent penetrance is why one evaluation never settles a relative's status.

· A dynamic murmur that grows when the ventricle empties separates obstructive hypertrophic cardiomyopathy from fixed valvular stenosis.

Common misconceptions

· Telling a genotype-positive adolescent with a normal echocardiogram that he is unaffected. Hypertrophy may still appear, and with MYBPC3 it can be delayed to middle age.

· Expecting a louder murmur with a fuller ventricle. A larger cavity moves the septum and leaflet apart and softens the murmur.

Where it is taught: Section C02, paragraphs on hemodynamics and on penetrance, and the stop and predict box of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C02-CA01

Clinical application: Collapse during a basketball game in a 17-year-old

A competent response contains

· Recognize: hypertrophic cardiomyopathy, autosomal dominant, most often from MYH7 encoding beta-myosin heavy chain or MYBPC3 encoding cardiac myosin-binding protein C.

· Explain: a mutant myosin head built into the thick filament beside normal myosin acts as a dominant negative, or a truncating MYBPC3 variant lowers protein through nonsense-mediated decay; heads are released from their folded resting state and thin filament variants raise Ca2+ sensitivity, so the heart is hypercontractile, wastes ATP, and relaxes poorly, and the energy deficit activates hypertrophic and fibrotic signaling that produces hypertrophy, myocyte disarray, fibrosis, and small vessel disease without any abnormal load.

· Differentiate: athlete's heart has an enlarged cavity with normal relaxation and regresses with detraining, while this boy has a small cavity and a 20 mm septum; hypertensive hypertrophy requires raised blood pressure, and his is 118/72 mm Hg; Fabry disease is X-linked from GLA; Friedreich ataxia brings ataxia and sensory loss; and Noonan syndrome brings the Turner-like features and pulmonary valve dysplasia.

· Investigate: the murmur becomes louder when ventricular volume falls on standing and softer with squatting or handgrip; the electrocardiogram shows high voltage with deep narrow Q waves from the thick septum; wall thickness of about 15 mm or more in an adult is diagnostic against about 11 mm or less normally; and tissue shows myocyte disarray with interstitial fibrosis and thick-walled intramural arteries.

· Decide: myocyte disarray and patchy interstitial fibrosis slow and scatter conduction and support reentry, so ventricular arrhythmia can be the first event, and relative risk depends on inheriting the variant, on age, and on modifiers, since each first-degree relative has a 50% chance of carrying it.

· Justify: standing lowers venous return and ventricular volume, bringing the septum and the anterior mitral leaflet closer during systole, so obstruction and turbulence increase, while squatting and handgrip raise volume or afterload and move them apart; relatives can carry the variant with a normal echocardiogram

because penetrance is incomplete and age dependent, which is why evaluation is repeated rather than done once.

Distinctions that matter

· Angina with normal coronary arteries here comes from high demand and narrow intramural vessels, not from epicardial obstruction.

· Sudden death in this disease comes from the arrhythmic substrate in the tissue, not from the outflow gradient itself.

Common misconceptions

· Reading normal left ventricular emptying as evidence against the diagnosis. The ventricle empties well and fails to fill.

· Calling a murmur that changes with posture a functional murmur. The change in loudness is the mechanism of the obstruction showing itself.

Where it is taught: Section C02, all paragraphs, the flow diagram, and the stop and predict box of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C03-BD1

Brain dump 1 · foundational retrieval: Marfan syndrome from gene to dissection

A competent response contains

· The aortic media is built from elastic lamellae with smooth muscle between them, held by collagen and microfibrils.

· Marfan syndrome affects about 1 in 5,000, about a quarter of cases are de novo, and it is autosomal dominant.

· The gene is FBN1 on chromosome 15 at 15q21, encoding fibrillin-1.

· Fibrillin-1 forms microfibrils that scaffold elastin and anchor the lens zonule, and microfibrils also sequester latent TGF-beta.

· Missense variants act as dominant negatives and truncating variants act by haploinsufficiency.

· Fibrillin deficiency increases TGF-beta signaling, matrix metalloproteinase activity, and matrix loss.

· Sequence: FBN1 variant with abnormal or reduced fibrillin-1, then weak microfibrils with excess TGF-beta signaling, then medial degeneration, then dilation of the root at the sinuses of Valsalva, then a larger radius raising wall tension, then aortic regurgitation followed by dissection or rupture.

· By Laplace's law wall tension rises with radius, so dilation feeds further dilation.

· The lesion is medial degeneration: elastic fiber fragmentation, smooth muscle loss, and proteoglycan pools; the older term cystic medial necrosis is misleading because there is neither a true cyst nor necrosis.

· The stretched annulus causes aortic regurgitation, and myxomatous mitral valves prolapse.

· Pleiotropy gives upward ectopia lentis, tall stature, arachnodactyly, pectus deformity, scoliosis, and joint laxity.

· Homocystinuria mimics the habitus but displaces the lens downward and causes thrombosis and intellectual disability.

· Ascending aortic dissection is the classic cause of death in untreated Marfan syndrome.

Distinctions that matter

· Upward against downward lens displacement separates Marfan syndrome from homocystinuria in a patient with the same body habitus.

· Fibrillin-1 has a structural job and a signaling job, and the aortic disease needs both the weakened scaffold and the excess TGF-beta.

Common misconceptions

· Taking cystic medial necrosis literally. There is no cyst and no necrosis, only fragmented elastin, lost smooth muscle, and pools of proteoglycan.

· Treating the root dilation as a fixed size. Laplace's law makes a wider root generate more wall tension, so the process accelerates.

Where it is taught: Section C03, first two paragraphs and the Marfan flow diagram of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C03-BD2

Brain dump 2 · integration and application: Three families, three failing wall components

A competent response contains

· Patient 1 is Marfan syndrome: FBN1, fibrillin-1, autosomal dominant, with weak microfibrils and excess TGF-beta producing medial degeneration worst at the aortic root, so the root dilates and the ascending aorta dissects.

· Patient 2 is Loeys-Dietz syndrome: usually TGFBR1 or TGFBR2, also SMAD3, TGFB2, and TGFB3, autosomal dominant, with aneurysms and dissections occurring younger, at smaller diameters, and throughout tortuous arteries.

· The Loeys-Dietz paradox is that the variants reduce receptor function in cells yet aortic TGF-beta signaling is increased.

· Patient 3 is vascular Ehlers-Danlos syndrome: COL3A1, type III collagen, autosomal dominant; type III collagen is a homotrimer abundant in arteries, bowel, uterus, and skin, so medium-sized arteries rupture or dissect without a preceding aneurysm and the sigmoid colon and gravid uterus can rupture.

· Most COL3A1 variants replace a glycine in the Gly-X-Y repeat, a position only glycine is small enough to fill, so the triple helix misfolds.

· In vascular Ehlers-Danlos syndrome the skin is thin and translucent but not markedly hyperextensible, and median survival is about 50 years.

· Arithmetic for a three-chain collagen: (1/2) to the third power is 1/8, so about 12.5% of molecules are fully normal with a glycine substitution, against about 50% of the usual amount all normal with a null allele.

· Arithmetic for a four-subunit channel such as KCNQ1: (1/2) to the fourth power is 1/16, about 6% working channels with a dominant negative subunit, against about 50% with a null allele.

· This is why dominant negative missense variants are often more severe than null alleles, and why null COL3A1 alleles cause later, milder disease.

· Clues outside the vessels: upward ectopia lentis with tall stature and arachnodactyly in Marfan; hypertelorism, bifid uvula or cleft palate, and craniosynostosis with no ectopia lentis in Loeys-Dietz; thin skin with bruising and colon and uterine rupture in vascular Ehlers-Danlos.

Distinctions that matter

· Rupture without a preceding aneurysm points to collagen failure, while progressive dilation before dissection points to loss of the elastic scaffold.

· Absence of ectopia lentis with hypertelorism and a bifid uvula separates Loeys-Dietz from Marfan syndrome, which share aortic disease and skeletal features.

Common misconceptions

· Assuming a null allele is always worse than a missense variant. For a multimer the mutant subunit spoils normal partners, so half the protein all normal beats one eighth of it fully normal.

· Grouping all three as the same disease because all three are autosomal dominant. The failing wall component predicts which vessels fail and how.

Where it is taught: Section C03, paragraphs on Loeys-Dietz and vascular Ehlers-Danlos, the comparison table, and the worked example of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C03-CP01

Comparison table: The three heritable disorders of the arterial wall

A competent response contains

· Gene and protein: FBN1 and fibrillin-1; TGFBR1 or TGFBR2 receptors, with SMAD3, TGFB2, and TGFB3 also reported; COL3A1 and type III collagen.

· All three are autosomal dominant.

· Arterial pattern: aortic root dilation with ascending dissection in Marfan; widespread aneurysms with tortuosity, younger and at smaller diameters, in Loeys-Dietz; rupture or dissection of medium-sized arteries in vascular Ehlers-Danlos.

· An aneurysm precedes the event in Marfan and Loeys-Dietz, and rupture occurs without a preceding aneurysm in vascular Ehlers-Danlos.

· Tissue level: medial degeneration with fragmented elastic fibers, lost smooth muscle, and proteoglycan pools in Marfan; reduced receptor function in cells with paradoxically increased aortic TGF-beta signaling in Loeys-Dietz; misfolded type III collagen helix from a glycine substitution in the Gly-X-Y repeat in vascular Ehlers-Danlos.

· Eye: upward ectopia lentis in Marfan; no ectopia lentis in Loeys-Dietz; no characteristic lens finding in vascular Ehlers-Danlos.

· Face, mouth, and skull: no characteristic findings listed for Marfan; hypertelorism, bifid uvula or cleft palate, and craniosynostosis in Loeys-Dietz; no characteristic findings listed for vascular Ehlers-Danlos.

· Skin, bowel, and uterus: no characteristic findings listed for Marfan or Loeys-Dietz; thin translucent skin that is not markedly hyperextensible, easy bruising, and sigmoid colon and gravid uterine rupture in vascular Ehlers-Danlos.

· Skeleton and joints: tall stature, arachnodactyly, pectus deformity, scoliosis, and joint laxity in Marfan; no separate skeletal list in the notes for the other two beyond the craniofacial findings in Loeys-Dietz.

· Median survival in vascular Ehlers-Danlos is about 50 years.

Distinctions that matter

· Marfan and Loeys-Dietz both dilate before they dissect, and the discriminator is age, diameter, how widespread the aneurysms are, and the craniofacial findings.

· The lens is the fastest separator between Marfan and Loeys-Dietz, and the skin, bowel, and uterus are the fastest separators for vascular Ehlers-Danlos.

Common misconceptions

· Waiting for a Marfan-sized aortic diameter in Loeys-Dietz. Dissection occurs at smaller diameters and younger ages.

· Expecting markedly hyperextensible skin in vascular Ehlers-Danlos. The skin is thin and translucent rather than stretchy.

Where it is taught: Section C03, comparison table on heritable disorders of the arterial wall of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C04-BD1

Brain dump 1 · foundational retrieval: Repolarizing currents and the long QT genotypes

A competent response contains

· Phase 3 repolarization depends on the rapid delayed rectifier current IKr from KCNH2, also called hERG, and the slow current IKs from KCNQ1 with its accessory subunit KCNE1; IK1 finishes repolarization.

· Repolarization reserve means the two currents share the work, so losing part of one may barely lengthen the resting QT interval.

· The four situations that unmask it are hypokalemia, bradycardia, a sympathetic surge, and an IKr-blocking drug, which is much of the explanation for incomplete penetrance.

· Congenital long QT syndrome affects about 1 in 2,000.

· LQT1 is the most common: KCNQ1 with loss of IKs, triggered by exercise and especially swimming, with a broad-based T wave.

· LQT2: KCNH2 with loss of IKr, triggered by startle and emotion with hypokalemia dangerous, and a low, notched or bifid T wave.

· LQT3: SCN5A gain of function producing a late Na+ current, with events at rest or in sleep, and a late T wave after a long isoelectric ST segment.

· The pattern across the three is that the K+ channel genes lose function while the Na+ channel gene gains function.

· Channelopathies cause sudden death in a structurally normal heart, so diagnosis rests on the electrocardiogram, family history, and genetic testing.

· Less outward K+ current or an added late inward Na+ current prolongs phases 2 and 3 and lengthens the QT interval.

Distinctions that matter

· IKr and IKs are different currents from different genes, and which one is lost predicts the trigger.

· Loss of function in a repolarizing current and gain of function in a depolarizing current reach the same long QT interval from opposite directions.

Common misconceptions

· Taking a normal resting QT interval as proof that a relative is unaffected. Repolarization reserve can hide a partial current loss until something else stresses it.

· Assuming every long QT gene loses function. SCN5A in LQT3 gains function by leaving a late Na+ current switched on.

Where it is taught: Section C04, first paragraph, the ventricular action potential diagram, and the long QT genotype table of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C04-BD2

Brain dump 2 · integration and application: Why the trigger, the ear, and the fever all follow from the current

A competent response contains

· Faints while swimming: LQT1, because IKs is increased by sympathetic stimulation and normally shortens the action potential during exercise, so losing it leaves repolarization unsupported exactly when it is needed.

· Collapses at a fire alarm: LQT2, because IKr dominates resting repolarization and is the current lost, and startle and emotion are the triggers; low extracellular K+ reduces IKr further, so hypokalemia is dangerous.

· Found dead in bed: LQT3, because a late Na+ current matters most at slow rates, so events occur at rest or in sleep.

· All three lead to early afterdepolarizations and torsades de pointes.

· Romano-Ward syndrome is the common autosomal dominant form with normal hearing.

· Jervell and Lange-Nielsen syndrome is autosomal recessive, from two abnormal alleles of KCNQ1 or KCNE1, and combines congenital sensorineural deafness with marked QT prolongation, often above 500 ms, and childhood sudden death.

· The stria vascularis secretes K+ into endolymph through KCNQ1 and KCNE1 channels to generate the endocochlear potential, so with no working channel hearing fails.

· One working allele suffices for the ear, so heterozygous parents hear normally but may have a mildly long QT interval.

· SCN5A shows allelic heterogeneity: gain of function causes LQT3, while loss of function causes Brugada syndrome and conduction disease, and some variants produce both.

Distinctions that matter

· The trigger reports which current is missing, because each current is recruited in a different physiological state.

· The ear needs only one working allele while the ventricle is affected by one bad allele, which is why the deafness is recessive and the long QT is dominant.

Common misconceptions

· Treating the triggers as coincidence. Exercise, startle, and rest each match the current that normally carries repolarization in that state.

· Assuming one gene gives one disease. SCN5A gives long QT when it gains function and Brugada syndrome when it loses function.

Where it is taught: Section C04, paragraphs on long QT triggers, Romano-Ward and Jervell and Lange-Nielsen, and the final paragraph on SCN5A of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C04-RT01

Retrieval exercise: Rebuild the Brugada sequence

A competent response contains

· Step 1: reduced peak Na+ current.

· Step 2: the transient outward current Ito deepens the epicardial phase 1 notch.

· Step 3: the epicardial dome is lost while the endocardium keeps its plateau.

· Step 4: the transmural gradient produces coved ST elevation in V1 and V2.

· Step 5: phase 2 reentry gives a closely coupled beat.

· Step 6: polymorphic ventricular tachycardia degenerates to ventricular fibrillation.

· The gene is SCN5A, loss of function, autosomal dominant with incomplete penetrance, found in only about 20% to 25% of patients.

· Right ventricular outflow epicardium has a large Ito and the endocardium does not, so a weaker upstroke lets Ito pull the epicardial membrane below L-type Ca2+ activation only there, which creates the transmural gradient.

· The type 1 electrocardiogram shows coved ST elevation of at least 2 mm in V1 or V2 running into a negative T wave.

· Fever, vagal tone, and Na+ channel blockers unmask the pattern; fever does it because some variant channels inactivate faster when warm, cutting Na+ current further.

· Events cluster at rest, in sleep, and with fever, and slowed outflow conduction probably contributes.

· It is most common in men and more often seen in Southeast Asia, and the larger male epicardial Ito helps explain the male predominance.

· Brugada syndrome causes ventricular fibrillation in a structurally normal heart.

Distinctions that matter

· The ST elevation here comes from a transmural repolarization gradient, not from the current of injury of ischemia.

· Loss of Na+ current is the first event and Ito is the current that exploits it, so the two are not interchangeable in the sequence.

Common misconceptions

· Expecting a positive gene test in most patients. SCN5A variants are found in only about a fifth to a quarter.

· Assuming fever is unrelated to the channel. Some variant channels inactivate faster when warm, so warming worsens the defect.

Where it is taught: Section C04, Brugada paragraph, flow diagram, and stop and predict box of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C04-PR01

Prediction exercise: Predict the effect of a low plasma potassium in LQT2

A competent response contains

· IKr falls, because the rapid delayed rectifier current drops when extracellular K+ is low.

· The margin left by repolarization reserve is used up, because her IKs cannot cover for an already reduced IKr.

· The ventricular action potential lengthens and the QT interval lengthens, because less outward K+ current prolongs phases 2 and 3.

· L-type Ca2+ channels have time to recover and reopen during the prolonged plateau.

· An early afterdepolarization appears, arising during repolarization in phases 2 and 3 from that inward Ca2+ current.

· The rhythm is torsades de pointes, and it can degenerate into ventricular fibrillation.

· Her risk rises further with an IKr-blocking drug, because it removes the same current that her variant and her low K+ have already reduced.

Distinctions that matter

· Low plasma K+ reduces IKr rather than increasing K+ efflux, which is why hypokalemia lengthens repolarization instead of shortening it.

· The early afterdepolarization starts the rhythm, and uneven repolarization across the wall is what sustains it.

Common misconceptions

· Expecting a normal baseline QT interval to protect her. Repolarization reserve only hides the defect until a second hit arrives.

· Treating the variant, the low K+, and the drug as three separate problems. All three subtract from the same current.

Where it is taught: Section C04, first paragraph on repolarization reserve and the LQT2 row of the genotype table of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C05-BD1

Brain dump 1 · foundational retrieval: Embryology to the characteristic heart defect

A competent response contains

· The endocardial cushions fuse to divide the atrioventricular canal, close the ostium primum, and form parts of the atrioventricular valves.

· Neural crest cells build the aorticopulmonary septum that divides the truncus arteriosus.

· Down syndrome is trisomy 21, about 95% from nondisjunction, mostly maternal meiosis I with advanced maternal age, about 4% from Robertsonian translocation, and about 1% from mosaicism.

· About 40% to 50% of patients with Down syndrome have a heart defect, classically an atrioventricular septal defect with an ostium primum defect, an inlet ventricular septal defect, and a common valve, from failed cushion fusion; other findings are hypotonia, a flat face, and duodenal atresia.

· Large shunts plus a tendency to early pulmonary vascular disease bring pulmonary hypertension sooner in Down syndrome.

· Turner syndrome is 45,X in about half of patients and otherwise mosaic or structural X changes, and it favors left-sided lesions: bicuspid aortic valve in about a fifth to a third, coarctation in about a tenth, and later aortic dilation and dissection; other findings are short stature, webbed neck, and ovarian insufficiency.

· One hypothesis in Turner syndrome is that fetal jugular lymphedema, which leaves the webbed neck, compresses the ascending aorta and reduces left heart flow.

· Coarctation causes arm hypertension, weak delayed femoral pulses, and rib notching from collaterals.

· The 22q11.2 deletion, called DiGeorge or velocardiofacial syndrome, occurs in about 1 in 4,000, removes about 3 million base pairs, is usually de novo, and is transmitted as autosomal dominant.

· Loss of one TBX1 copy disturbs the pharyngeal arches and the neural crest, giving conotruncal defects: interrupted aortic arch type B, truncus arteriosus, tetralogy of Fallot, and a right-sided arch.

· Third and fourth pouch failure causes thymic hypoplasia with T cell deficiency and parathyroid hypoplasia with hypocalcemia, summarized as CATCH-22, along with cleft palate.

· Williams syndrome is a 7q11.23 microdeletion including ELN; elastin also tells smooth muscle to stop proliferating, so haploinsufficiency thickens the media, causing supravalvular aortic stenosis at the sinotubular junction and peripheral pulmonary and renal artery stenosis, with elfin facies, hypercalcemia, and a friendly personality.

· Noonan syndrome is autosomal dominant through the RAS pathway, often PTPN11, with dysplastic pulmonary valve stenosis and hypertrophic cardiomyopathy, Turner-like features, a normal karyotype, and involvement of both sexes.

Distinctions that matter

· Cushion disease gives central atrioventricular defects, while neural crest disease gives outflow and arch defects, which is why Down syndrome and the 22q11.2 deletion have different lesions.

· Turner syndrome and Noonan syndrome share the physical habitus but differ in karyotype, in which side of the heart is affected, and in which sexes are involved.

Common misconceptions

· Explaining the Williams arterial lesion as a missing structural protein alone. Elastin also signals smooth muscle to stop proliferating, so losing one copy thickens the media.

· Expecting a ventricular septal defect as the classic Down syndrome lesion. The classic lesion is a complete atrioventricular septal defect from failed cushion fusion.

Where it is taught: Section C05, opening paragraph and the paragraphs on each syndrome of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C05-BD2

Brain dump 2 · integration and application: Read the findings back to the mechanism

A competent response contains

· Child 1 is Down syndrome, trisomy 21, with an atrioventricular septal defect from failed endocardial cushion fusion leaving a common atrioventricular valve.

· Child 1 reaches pulmonary hypertension sooner because a large shunt is combined with a tendency to early pulmonary vascular disease.

· Child 2 is the 22q11.2 deletion, with loss of one TBX1 copy disturbing the pharyngeal arches and neural crest to give interrupted aortic arch type B.

· In child 2 the immune and parathyroid findings travel with the cardiac findings because the third and fourth pharyngeal pouches fail alongside the arch and crest derivatives, giving thymic hypoplasia with T cell deficiency and parathyroid hypoplasia with hypocalcemia.

· Child 3 is Turner syndrome, 45,X or mosaic, with coarctation; the lesion gives arm hypertension, weak delayed femoral pulses, and rib notching from collaterals, and her aorta is at risk of later dilation and dissection.

· Child 4 is Williams syndrome, a 7q11.23 microdeletion including ELN, with supravalvular aortic stenosis at the sinotubular junction from a thickened media.

· In child 4 an ELN variant alone causes only familial supravalvular aortic stenosis, because the neighboring genes in the deletion account for the facies, the calcium abnormality, and the personality.

· Child 5 is Noonan syndrome, autosomal dominant through the RAS pathway and often PTPN11, with dysplastic pulmonary valve stenosis and hypertrophic cardiomyopathy and a normal karyotype.

· A syndrome has one cause, an association is a nonrandom clustering without a single cause, and a sequence is a cascade from one defect.

· Examples from the notes: Down syndrome as a syndrome, VACTERL as an association, and Potter sequence as a sequence.

· VACTERL needs at least three features, is usually sporadic with low recurrence risk, and reflects an early mesodermal disturbance, most often with a ventricular septal defect.

· Holt-Oram syndrome from TBX5 gives septal defects with upper limb anomalies, and Fanconi anemia gives radial defects with marrow failure.

Distinctions that matter

· An association names a pattern and not a cause, which is why its recurrence risk is low and no single test confirms it.

· Radial or upper limb defects with a septal defect should raise a single-gene look-alike rather than stopping at the association.

Common misconceptions

· Treating VACTERL as a diagnosis with a genetic test. It is a clustering of features with no single cause.

· Reading hypocalcemia and infection in a child with an arch anomaly as unrelated problems. They come from the same pouch failure.

Where it is taught: Section C05, paragraphs on each syndrome, the paragraph on VACTERL, and the syndrome table of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C05-CP01

Comparison table: Genetic syndromes and their cardiovascular lesions

A competent response contains

· Down syndrome: trisomy 21; atrioventricular septal defect; failed endocardial cushion fusion; hypotonia, flat face, duodenal atresia.

· Turner syndrome: 45,X or mosaic; bicuspid aortic valve and coarctation; reduced left heart flow, with fetal jugular lymphedema compressing the ascending aorta as one hypothesis; short stature, webbed neck, ovarian insufficiency.

· 22q11.2 deletion syndrome: microdeletion of about 3 million base pairs including TBX1; interrupted arch type B, truncus arteriosus, tetralogy of Fallot; disturbed pharyngeal arches and neural crest; thymic hypoplasia, hypocalcemia, cleft palate.

· Williams syndrome: 7q11.23 deletion including ELN; supravalvular aortic stenosis with peripheral pulmonary and renal artery stenosis; elastin haploinsufficiency allowing smooth muscle proliferation and a thickened media; elfin facies, hypercalcemia, friendly personality.

· VACTERL: a sporadic association with no single cause; ventricular septal defect most often; early mesodermal disturbance; vertebral, anal, tracheoesophageal, renal, and radial limb defects.

· Noonan syndrome: autosomal dominant RAS pathway, often PTPN11; dysplastic pulmonary valve stenosis and hypertrophic cardiomyopathy; RAS pathway signaling in valve and myocardial development; Turner-like features with a normal karyotype in both sexes.

Distinctions that matter

· Down syndrome and the 22q11.2 deletion sit at opposite ends of the heart: cushions in the middle against outflow and arch at the exit.

· Turner syndrome and Williams syndrome both narrow the left-sided outflow, but one is a valve and arch problem and the other is a thickened media above the valve.

Common misconceptions

· Filling the Noonan row with Turner karyotype findings. The karyotype is normal and both sexes are affected.

· Giving VACTERL a genetic basis in the second column. It is an association, so the entry is that it is sporadic with no single cause.

Where it is taught: Section C05, table of genetic syndromes and their characteristic cardiovascular lesions of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C05-Q01

Board-style question: A newborn with hypocalcemia and an arch anomaly

Correct answer: C. Loss of one copy of TBX1 disturbing the pharyngeal arches and the neural crest

Why it is right: The deletion removes one copy of TBX1, which disturbs the pharyngeal arches and the neural crest that build the aorticopulmonary septum and the arch, giving interrupted aortic arch type B, while failure of the third and fourth pharyngeal pouches produces thymic hypoplasia with T cell deficiency and parathyroid hypoplasia with hypocalcemia.

A is wrong: Trisomy 21 with failed cushion fusion gives an atrioventricular septal defect with a common valve, together with hypotonia, a flat face, and duodenal atresia, and it does not explain the absent thymus or the low calcium.

B is wrong: Elastin haploinsufficiency in Williams syndrome thickens the arterial media and gives supravalvular aortic stenosis with hypercalcemia rather than an interrupted arch with hypocalcemia.

D is wrong: A GAA repeat expansion in FXN lowers frataxin and causes Friedreich ataxia with concentric hypertrophy later in childhood, not a neonatal arch interruption with thymic and parathyroid failure.

A competent response contains

· Hypocalcemia, absent thymus, cleft palate, and interrupted aortic arch type B are the 22q11.2 deletion.

· Loss of one TBX1 copy disturbs the pharyngeal arches and the neural crest, and failure of the third and fourth pouches explains the thymic and parathyroid findings.

Distinctions that matter

· Conotruncal and arch defects follow neural crest disease, while central atrioventricular defects follow cushion disease.

Common misconceptions

· Treating the cardiac, immune, and endocrine findings as separate conditions in one infant. They share one developmental origin.

Where it is taught: Section C05 of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C06-BD1

Brain dump 1 · foundational retrieval: Dystrophin, frataxin, and mitochondrial DNA

A competent response contains

· Dystrophin links actin through beta-dystroglycan and alpha-dystroglycan to laminin in the basement membrane, protecting the sarcolemma during contraction.

· DMD sits at Xp21 and is the largest human gene, so about a third of cases are new variants; Duchenne and Becker dystrophies are X-linked recessive.

· Duchenne dystrophy affects about 1 in 3,500 to 5,000 males, and deletions make up about two thirds of variants.

· Reading frame rule: in-frame deletions make a shortened, partly functional dystrophin, which is Becker, and frameshifts create a premature stop with no dystrophin, which is Duchenne; the rule is correct in about 90% of cases.

· Without dystrophin, contraction tears the membrane, Ca2+ enters, and proteases and mitochondrial Ca2+ overload cause necrosis.

· Creatine kinase is very high, with Gowers sign and calf pseudohypertrophy.

· Cardiac fibrosis starts subepicardially in the inferolateral left ventricle, producing dilated cardiomyopathy by the late teens, with tall R waves in V1, lateral Q waves, and sinus tachycardia; Becker cardiomyopathy can outpace limb weakness.

· Female carriers can develop cardiomyopathy when random or skewed X inactivation leaves many myocytes expressing only the mutant allele.

· Friedreich ataxia is autosomal recessive from a GAA repeat expansion in the first intron of FXN, normal up to about 33 repeats and disease alleles about 66 to more than 1,000; the expansion silences transcription.

· Frataxin is needed for mitochondrial iron-sulfur cluster assembly, which aconitase and complexes I to III depend on, so energy failure, mitochondrial iron accumulation, and oxidative damage hit dorsal root ganglia and spinocerebellar tracts, the heart, and beta cells.

· Friedreich cardiomyopathy is concentric hypertrophy without extensive disarray, with later fibrosis and dilation, and is the usual cause of death.

· Mitochondrial DNA is about 16,600 base pairs and encodes 13 respiratory chain subunits, 22 transfer RNAs, and 2 ribosomal RNAs, and is maternally inherited.

· Heteroplasmy, random segregation, and the oocyte bottleneck make mutant fractions vary between tissues and siblings, and disease appears where a tissue's threshold is exceeded, lowest in brain, retina, muscle, and heart.

· Kearns-Sayre syndrome is a large, usually sporadic mitochondrial DNA deletion with onset before age 20, giving progressive external ophthalmoplegia, pigmentary retinopathy, and conduction block, high CSF protein, or ataxia; His-Purkinje block can progress to complete heart block and muscle shows ragged red fibers.

Distinctions that matter

· Deletion size does not predict severity; the reading frame does.

· Membrane fragility causes dilation, energy failure causes hypertrophy, and a mitochondrial DNA deletion favors the conduction system.

Common misconceptions

· Assuming a larger deletion means Duchenne. A large in-frame deletion still makes a partly working protein and gives Becker dystrophy.

· Treating Friedreich ataxia as a coding variant. The GAA expansion sits in the first intron and silences transcription rather than altering the protein sequence.

Where it is taught: Section C06, dystrophin paragraphs, worked example, Friedreich paragraph, mitochondrial paragraph, and the disorder table of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C06-BD2

Brain dump 2 · integration and application: Work the frame, the carrier, and the threshold

A competent response contains

· Boy A: 522 divided by 3 is 174 with no remainder, so 174 codons are removed, the downstream frame is intact, a shortened dystrophin with both binding ends is made, and the disease is Becker.

· Boy B: 197 divided by 3 leaves a remainder of 2, so the frame shifts, a premature stop appears, and no dystrophin is made, which is Duchenne.

· Boy A has the larger deletion and the milder disease, because the reading frame rather than the amount of sequence lost decides whether a protein is made.

· A premature stop codon causes the transcript to undergo nonsense-mediated decay.

· The carrier develops cardiomyopathy because random X inactivation leaves some myocytes expressing only the mutant DMD allele, and if enough do, through chance or skewed inactivation, those cells are damaged with each contraction.

· Her heart cannot replace them because cardiomyocytes are permanent cells, so the damaged myocytes are replaced by fibrosis.

· The two sisters differ because heteroplasmy, random segregation, and the oocyte bottleneck give them different mutant fractions, and disease appears only where a tissue's threshold is exceeded.

· Thresholds are lowest in brain, retina, muscle, and heart, because those tissues depend most heavily on oxidative energy supply.

· Membrane fragility gives a dilated ventricle because repeated myocyte loss and fibrosis leave a weak wall that stretches.

· Failure of energy supply gives a thick ventricle, as concentric hypertrophy in Friedreich ataxia.

· A large mitochondrial DNA deletion shows itself in the conduction system as progressive His-Purkinje block that can reach complete heart block.

Distinctions that matter

· The reading frame rule is arithmetic on nucleotides, not a judgment about deletion size, and it holds in about 90% of cases.

· X inactivation and heteroplasmy are different mechanisms that produce the same result, which is cell to cell variation within one person.

Common misconceptions

· Calling a female carrier of an X-linked recessive disease immune to the phenotype. Enough myocytes expressing only the mutant allele produce cardiomyopathy.

· Expecting siblings who inherit the same mitochondrial variant to be equally affected. The mutant fraction they receive and keep differs.

Where it is taught: Section C06, worked example on the reading frame rule, the carrier sentence, the mitochondrial paragraph, and the stop and predict box of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C06-Q02

Board-style question: Progressive block in a young adult

Correct answer: A. A large deletion of mitochondrial DNA removing respiratory chain subunits and transfer RNAs

Why it is right: A large mitochondrial DNA deletion removes respiratory chain subunits and transfer RNAs, so oxidative energy supply fails in the tissues with the lowest threshold, giving progressive external ophthalmoplegia, pigmentary retinopathy, and His-Purkinje conduction block before age 20, with ragged red fibers on muscle biopsy, and the deletion is usually sporadic so relatives are unaffected.

B is wrong: A frameshift DMD deletion abolishing dystrophin causes Duchenne dystrophy in males, with very high creatine kinase, calf pseudohypertrophy, and dilated cardiomyopathy from inferolateral fibrosis, not ophthalmoplegia with retinopathy.

C is wrong: A GAA repeat expansion in FXN causes Friedreich ataxia with ataxia, sensory loss, and concentric hypertrophy rather than progressive external ophthalmoplegia and ragged red fibers.

D is wrong: A dominant negative MYH7 variant causes hypertrophic cardiomyopathy with myocyte disarray and a thick nondilated ventricle, and it does not affect the eyes, the retina, or skeletal muscle histology in this way.

A competent response contains

· Ophthalmoplegia with ptosis, pigmentary retinopathy, and progressive conduction block before age 20 is Kearns-Sayre syndrome.

· Ragged red fibers point to a mitochondrial DNA disorder, and the lesion is a large, usually sporadic deletion, which is why the mother and brothers are unaffected.

Distinctions that matter

· A large mitochondrial DNA deletion favors the His-Purkinje system, which separates it from the dilated and hypertrophic phenotypes of the other inherited myocardial diseases.

Common misconceptions

· Expecting every mitochondrial disorder to show maternal transmission. A large deletion of this kind is usually sporadic.

Where it is taught: Section C06 of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C07-BD1

Brain dump 1 · foundational retrieval: Why the adult heart grows by hypertrophy

A competent response contains

· Fetal cardiomyocytes divide, but around birth they largely leave the cell cycle, and many become polyploid or binucleate.

· Cell cycle exit is maintained by low cyclin and cyclin-dependent kinase activity, the inhibitors p21 and p27, and retinoblastoma protein repression of E2F.

· Carbon-14 dating shows renewal of only about 1% per year in young adults, and no confirmed adult cardiac stem cell rebuilds muscle, so the heart is a permanent tissue.

· Skeletal muscle regenerates from satellite cells and smooth muscle can divide, but cardiac muscle has no such reserve.

· By Laplace's law, ventricular wall stress is proportional to pressure times radius divided by twice the wall thickness.

· Pressure overload from hypertension or aortic stenosis adds sarcomeres in parallel and thickens the wall, which is concentric hypertrophy; volume overload from regurgitation adds sarcomeres in series and enlarges the cavity, which is eccentric hypertrophy; both tend to return wall stress to normal.

· Normal heart weight is about 250 to 300 g in women and about 300 to 350 g in men.

· Stretch is sensed by integrins, Z disc proteins, and stretch-activated channels.

· Angiotensin II, endothelin-1, and norepinephrine act through Gq, phospholipase C, Ca2+, and protein kinase C, and sustained Ca2+ activates calcineurin, which dephosphorylates NFAT so it enters the nucleus and activates hypertrophic genes with GATA4 and MEF2.

· Exercise hypertrophy uses IGF-1 through PI3K and Akt instead.

· The fetal gene program raises ANP in the ventricle, BNP in proportion to stretch, beta-myosin, and skeletal alpha-actin, and lowers alpha-myosin and SERCA2a, with a shift toward glucose use.

· Consequences: slower myosin saves energy, less SERCA2a slows relaxation, and the natriuretic peptides raise cGMP to oppose the renin-angiotensin-aldosterone system.

· BNP below about 100 pg/mL makes heart failure an unlikely cause of acute dyspnea.

Distinctions that matter

· Concentric and eccentric hypertrophy differ in where sarcomeres are added, which is why one thickens the wall and the other enlarges the cavity.

· The IGF-1 route through PI3K and Akt and the Gq route through calcineurin and NFAT both enlarge myocytes, and the route decides whether the fetal gene program switches on.

Common misconceptions

· Describing adult cardiac hypertrophy as an increase in cell number. The myocytes enlarge, since they have left the cell cycle.

· Reading a raised BNP as a hormone that worsens the picture. It raises cGMP and opposes the renin-angiotensin-aldosterone system, and it is useful as a measure of stretch.

Where it is taught: Section C07, paragraphs on cell cycle exit, Laplace's law, signaling, and the comparison table of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C07-BD2

Brain dump 2 · integration and application: Two thick hearts and one scar

A competent response contains

· The rower: IGF-1 through PI3K and Akt, the fetal gene program not activated, capillaries keeping pace with no fibrosis, and regression when training stops.

· The hypertensive man: Gq with calcineurin and NFAT, the fetal gene program activated with ANP, BNP, and beta-myosin, capillaries lagging with interstitial fibrosis, and hypertrophy that may progress to failure.

· Pathologic hypertrophy progresses because capillaries lag, SERCA2a falls, and collagen accumulates, so ischemia and mishandled Ca2+ kill myocytes.

· The ventricle dilates despite starting thick because lost myocytes are not replaced, the survivors are overloaded, and a larger radius raises wall stress again.

· Repair after infarction: macrophages clear dead tissue and release TGF-beta, fibroblasts convert into contractile myofibroblasts, type III collagen is laid down, and it is later replaced by cross-linked type I collagen.

· The scar is strong but cannot contract and cannot conduct, so surviving myocytes hypertrophy to carry the load and border-zone strands support reentrant ventricular tachycardia.

· Stunned and hibernating myocardium can recover; scar cannot.

· Every disease here that kills myocytes ends in fibrosis, hypertrophy of the survivors, and an arrhythmic substrate, because cardiomyocytes are permanent cells.

· The same repair follows myocyte loss in the muscular dystrophies and in Friedreich ataxia.

Distinctions that matter

· Physiologic and pathologic hypertrophy both enlarge existing myocytes, and the signaling route rather than the amount of thickening decides the outcome.

· Recoverable myocardium and scar look alike on a wall motion study and differ entirely in whether the cells are alive.

Common misconceptions

· Reading dilation as a different disease from the earlier thickening. It is the same cycle after enough myocytes have been lost.

· Treating scar as inert. It cannot contract or conduct, and its border zone is where reentrant ventricular tachycardia arises.

Where it is taught: Section C07, comparison table, maladaptive cycle, and the final paragraph on repair of the CV-14 Cardiovascular Genetics notes.

NPLEX-P1-CARD-14-01-C07-RT01

Retrieval exercise: Rebuild the maladaptive hypertrophy cycle

A competent response contains

· Step 1: sustained load raises wall stress.

· Step 2: myocytes hypertrophy and fetal genes switch on.

· Step 3: capillaries lag, SERCA2a falls, and collagen accumulates.

· Step 4: ischemia and Ca2+ mishandling kill myocytes.

· Step 5: the survivors are overloaded and the ventricle dilates.

· Step 6: a larger radius raises wall stress again, closing the loop.

· The genes switched on are ANP in the ventricle, BNP in proportion to stretch, beta-myosin heavy chain, and skeletal alpha-actin, with alpha-myosin and SERCA2a switched down.

· Less SERCA2a slows relaxation, and lagging capillaries with accumulating collagen reduce oxygen supply to a thicker wall.

· The survivors have more work because the lost myocytes are gone and the same load is shared among fewer cells.

· By Laplace's law wall stress is proportional to pressure times radius divided by twice wall thickness, so a dilated cavity raises wall stress and sends the loop round again.

· Renewal is only about 1% per year and no confirmed adult cardiac stem cell rebuilds muscle, so lost myocytes are replaced by scar and the loop cannot be broken by replacing cells.

Distinctions that matter

· Hypertrophy at first lowers wall stress, and it is the later loss of myocytes and the dilation that raise it again.

· The fetal gene program is an adaptation that saves energy and slows relaxation at the same time, so it is neither purely protective nor purely harmful.

Common misconceptions

· Drawing the cycle as a straight line ending in failure. Each turn regenerates the stimulus that started it.

· Expecting new myocytes to replace the lost ones over time. Renewal of about 1% per year cannot keep pace.

Where it is taught: Section C07, cycle diagram on maladaptive pathologic hypertrophy of the CV-14 Cardiovascular Genetics notes.